INCH-POUND

MIL-R-11521G
29 October 1993
SUPERSEDING
MIL-R-11521F
8 October 1980

MILITARY SPECIFICATION

RINGS, EXPANSION, HOSE, BRASS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers brass expansion rings for use with hose couplings.

1.2 Classification. Rings shall be the sizes and ring lengths shown in table I, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 4730

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-2073-1	- DoD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-2073-2	- Packaging Requirements Code.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

* 2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 135	- Seamless Brass Tube.
B 154	- Mercurous Nitrate Test for Copper and Copper Alloys.

(Application for copies should be addressed to the AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Description. The expansion rings shall be as shown in figure 1 and as specified herein.

* 3.2 Material. The rings shall be soft-annealed seamless brass tube conforming to ASTM B 135, copper alloy UNS no. 023000, temper 06.

3.2.1 Mechanical properties.

3.2.1.1 Expansion. When tested as specified in ASTM B 135, the tubing from which the rings are fabricated shall not rupture or crack.

3.2.1.2 Residual stress. When tested as specified in ASTM B 154, the tubing from which the rings are fabricated shall not show evidence of stress-corrosion cracking caused by residual stress.

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3.3 Dimensions and tolerances. The nominal hose size and length of the rings shall be as specified (see 6.2). The outside diameter, length, and wall thickness shall be as shown in table I. The ends of the rings shall be perpendicular to the centerline within ±0.016 inch. The letters "(A)", "(B)", and "(C)", in column headings of table I refer to the corresponding letters in figure 1.

* TABLE I. Sizes and Dimensions.

Nominal hose size	Ring outside diameter (A)	Ring length +1/32, -0 (B)	Wall Thickness (C)
3/4	0.750 ±0.003	3/4, 1-1/4, 1-3/8	0.049 ±.004
1	1.000 ±0.003	7/8, 1, 1-1/4	0.049 ±.004
1-1/2	1.500 ±0.003	3/4, 1-1/4, 1-3/8	0.049 ±.004
2	2.000 ±0.003	1, 1-3/8	0.049 ±.004
2-1/2	2.500 ±0.004	1-1/4, 1-1/2	0.065 ±.006
3	3.000 ±0.004	1-7/8	0.065 ±.006
4	4.000 ±0.005	2	0.083 ±.006

Note: All dimensions are in inches, see figure 1.

3.4 Finish. Annealed tubing from which the rings are fabricated shall be either bright annealed or acid cleaned after final annealing operations.

* 3.5 Identification marking. The rings shall be identified in accordance with MIL-STD-130. The part number shall be marked on the bare item and shall be the specification part number or manufacturer's commercial part number.

3.6 Workmanship. The rings shall be free from scale, grooving, indentations, cracks, scores, dents, and burrs. The ends shall be rounded and smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of all rings of the same size offered for delivery at one time.

* 4.3.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, using table I and IIa, inspection level I for examination and inspection level S2 for tests. A lot is accepted when 0 defects are present and rejected when 1 or more defects are found.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined for defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. Failure to pass any test shall be causes for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The rings shall be examined as specified herein for the following defects:

Major

101. Material not as specified (see 3.2).
102. Outside diameter not as specified (see 3.3 and 6.2).
103. Length not as specified (see 3.3 and 6.2).
104. Wall thickness not as specified (see 3.3).
105. Ends of ring not perpendicular to the centerline (see 3.3).
106. Finish not as specified (see 3.4).
107. Identification marking missing or illegible (see 3.5).
108. Workmanship not as specified (see 3.6).

4.4.2 Tests. The rings shall be subjected to the following tests.

4.4.2.1 Expansion. The ring shall be tested in accordance with ASTM B 135. Visible evidence of cracking or rupture shall constitute failure of this test.

4.4.2.2 Mercurous nitrate. The rings shall be tested in accordance with ASTM B 154. Visible evidence of cracking shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and marking shall be in accordance with the quality assurance provisions in MIL-H-775.

* 5. PACKAGING

* 5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as required by the contracting activity in accordance with MIL-STD-2073 and MIL-STD-129.

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Expansion rings are used to secure hose and couplings together.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Nominal hose size and ring length required (see table II).
c. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. PIN number marked on ring (see 3.5).
e. Level of preservation and of packing required (see 5.1).

* 6.3 Part or identifying system. The part or identifying number (PIN) of rings covered by this specification should be designed in the following form (see 3.5 and 6.2).

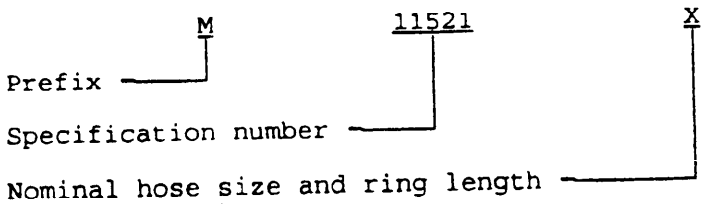


TABLE II. Nominal Hose Size and Ring Length.

	Ring length +1/32 -0							
	3/4	7/8	1	1-1/4	1-3/8	1-1/2	1-7/8	2
3/4	01	03	04	05	08	11	12	13
1	02			06	09			14
1-1/2				07	10			15
2								
2-1/2								
3								
4								
4-1/2								
5								

*

6.4 Subject term (key word) listing.

Clamp, hose
Coupling, hose
Clamp, brass
Sleeve, brass

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - SH
Air Force - 99

Review activities:

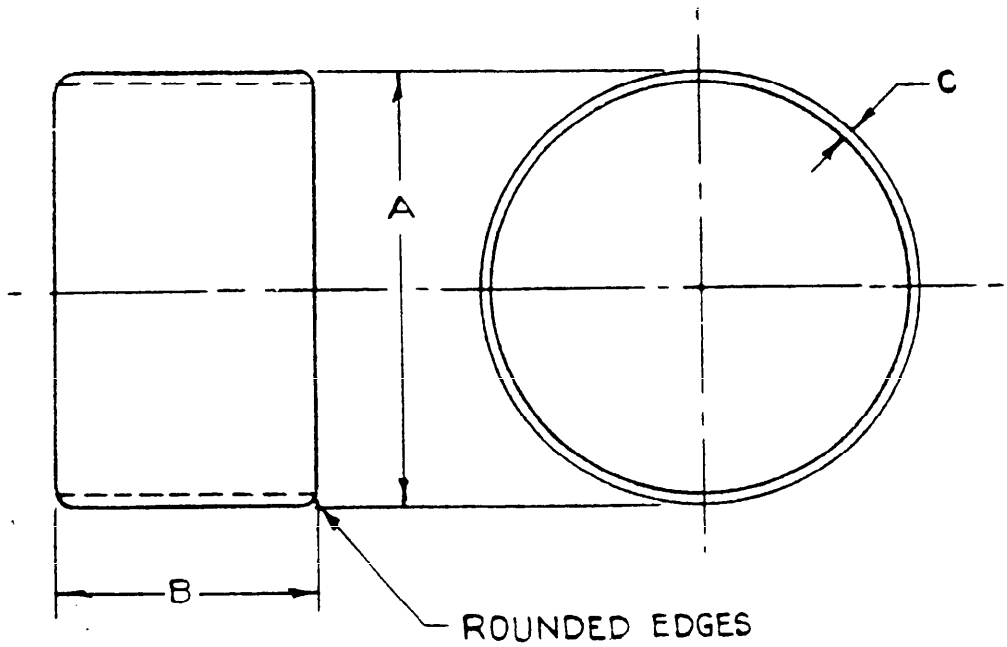
Air Force - 82
DLA - CS

Preparing activity:

Army - ME

Project 4730-0122

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LEGEND
A - OUTSIDE DIAMETER
B - LENGTH
C - THICKNESS

FIGURE 1. Expansion ring.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-11521G	2. DOCUMENT DATE (YYMMDD) 931029

3. DOCUMENT TITLE
Rings, Expansion, Hose, Brass

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340	

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